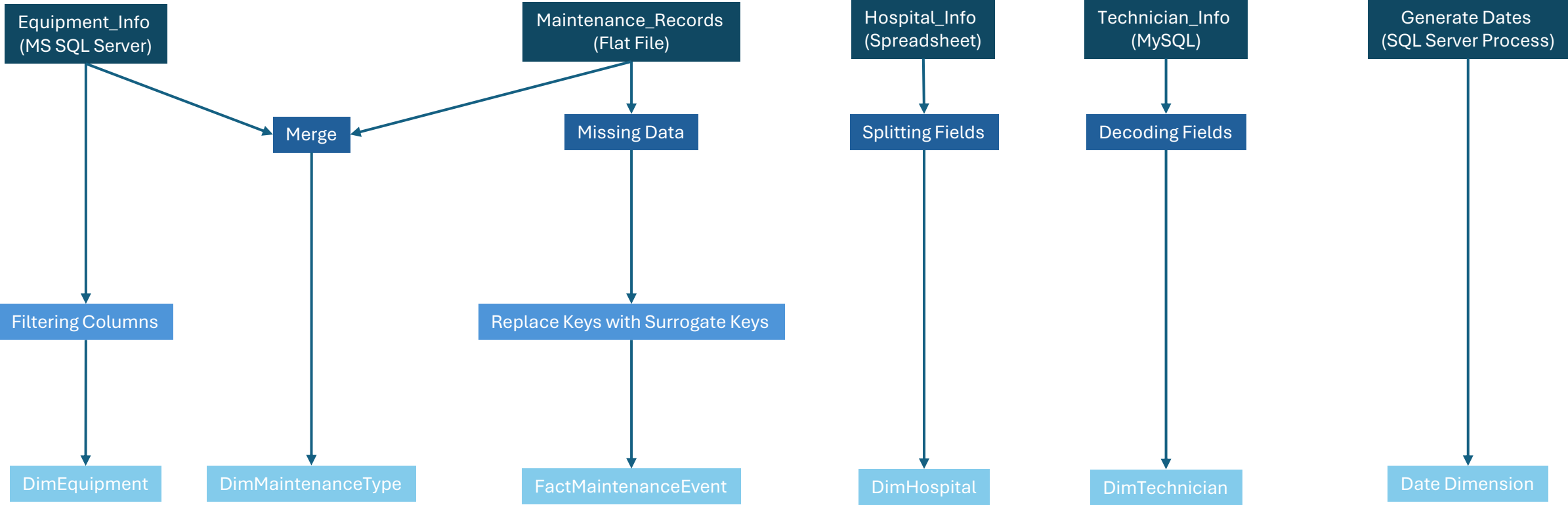


# Data Staging



**Decoding Fields:** Convert coded values in the Technician data (such as certification codes 0/1) into meaningful, human-readable values.

**Merge (Maintenance Type):** Combine Equipment data and Maintenance Records using a common attribute (such as Equipment Type) to create the DimMaintenanceType table.

**Filtering Columns (Equipment):** Remove unnecessary attributes from the Equipment data that are not required in the DimEquipment table.

**Missing Data:** Identify and handle missing values in Maintenance Records by either removing incomplete records or replacing missing values with defaults (such as 0).

**Replace Operational Keys with Surrogate Keys:** Replace source system identifiers (such as TechnicianID, HospitalID, and EquipmentID) with surrogate keys from the dimension tables before loading into the FactMaintenanceEvent table.

**Splitting Fields:** Split combined fields in the Hospital data (such as location) into separate attributes like City and State for the DimHospital table.

**Date Generation:** Generate date values to populate the DimDate table, ensuring consistent time-based analysis.